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Crude oil Price forecasting: Leveraging machine learning for global economic stability☆

Amar Rao [a], Gagan Deep Sharma [b,f], Aviral Kumar Tiwari [c,*], Mohammad Razib Hossain [d,e], Dhairya Dev [b]

[a] BML Munjal University, 67th KM Stone, NH-8, Gurugram, Haryana-122413, India
[b] University School of Management Studies, Guru Gobind Singh Indraprastha University, Sector 16C, Dwarka, New Delhi 110078, India
[c] Indian Institute of Management Bodh Gaya (IIM Bodh Gaya), Bodh Gaya 824234, Bihar, India
[d] School of Economics and Public Policy, Adelaide Business School, The University of Adelaide, Australia
[e] Department of Agricultural Finance and Cooperatives, Bangabandhu Sheikh Mujibur Rahman Agricultural University, Gazipur, Bangladesh
[f] Centre of International Programmes, Széchenyi István Egyetem, Győr, Hungary

A R T I C L E I N F O

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A B S T R A C T

The volatility of the energy market, particularly crude oil, significantly impacts macroeconomic indices, such as inflation, economic growth, currency exchange rates, and trade balances. Accurate crude oil price forecasting is crucial to risk management and global economic stability. This study examines various models, including GARCH (1,1), Vanilla LSTM, GARCH (1,1) LSTM, and GARCH (1,1) GRU, to predict Brent crude oil prices using different time frequencies and sample periods. The LSTM and GARCH (1,1)-GRU hybrid models showed superior performance, with LSTM slightly better in predictive accuracy and GARCH (1,1)-GRU in minimizing squared errors. These findings emphasize the importance of precise crude oil price forecasting for the global energy market and manufacturing sectors that rely on crude oil prices. Accurate forecasting helps ensure economic sustainability and stability and prevents disruptions to production and distribution chains in both developed and emerging economies. Policymakers may choose to implement energy security measures in response to the significant impact of crude oil price volatility on the macroeconomic indicators. These measures could include maintaining strategic reserves, diversifying energy sources, and decreasing the dependence on volatile oil markets. By doing so, a country's ability to handle oil price fluctuations and ensure a stable energy supply can be enhanced.

1. Introduction

Fossil energy has been the building block of the world's economic system since the Industrial Revolution. Economic sustainability and its attainment largely confide the successful co-movement of the expansion of the energy market and manufacturing sectors of a particular economy. For the successful expansion and functioning of the global energy market, all available agents must function perfectly, which is crucial to avoid market failure. Since fossil fuels (i.e., oil, natural gas, and coal) are raw materials with irredeemable influences, market failure, uncertainty, and other negative externalities in the fossil fuel market can severely affect the global economic system through price volatility(Wang et al., 2022). Among the few available categories of fossil energy, oil has been the most traded fossil fuel on the planet, which indicates the relative importance of crude oil as an input in the production process(Heo et al., 2010; Zhang et al., 2020; Zhao et al., 2016). According to the European Commission's estimation (2023), among the different types of fossil energy traded globally in 2015, roughly 60 % were oil, and 25 % were natural gas.

Oil and gas are less appealing energy options than nuclear and renewable energy sources. However, these old-school fossil energies have less cumulative consumption-induced environmental costs than other dirty avenues of fossil energy, such as coal(Zhou et al., 2021). The number of available substitutes for crude oil is minimal, given that other close substitutes (i.e., coal) have a higher pollution intensity, which forces the demand for this invaluable input to be very inelastic. In addition, compared to the bottleneck number of available oil fields, the global cumulative demand for crude oil surges exponentially, leaving a

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* Corresponding author at: Indian Institute of Management Bodh Gaya (IIM Bodh Gaya), Bodh Gaya 824234, Bihar, India.
E-mail addresses: amarydvrao@gmail.com (A. Rao), aviral.t@iimbg.ac.in (A.K. Tiwari), mohammadrazib.hossain@adelaide.edu.au (M.R. Hossain).

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footprint of market asymmetry, followed by a colossal gap in the crude oil supply chain in the energy industry. According to a recent forecast of global demand for crude oil in 2025, the Organization of the Petroleum Exporting Countries (OPEC, 2023a, 2023b) postulates that the cumulative demand for crude oil will hit a new record of 113 million barrels/day (mb/d) by 2025, which further substantiates that the consumption of crude oil has surged by an average of 30 million barrels/day (mb/d) between 2000 and 2025. The forecasted cumulative demand for crude oil was gauged by considering the post-COVID-19 pandemic-related effects. Most recent studies have noted that the post-pandemic consumption of fossil fuels is on the rise, and in some nations, the consumption of fossil fuels has surpassed the pre-pandemic level(Davis et al., 2022; Le Billon et al., 2021; Smith et al., 2021). Because energy depletion is inevitable(Hossain et al., 2023), it can unleash an energy market that can be more volatile in the future due to negative environmental externalities, market asymmetry, structural changes in the economy, and unregulated market conditions(Hailemariam and Smyth, 2019; Wang et al., 2022). Given that the economic performance of many developed and emerging developing economies predominantly confides in the stability of crude oil prices, volatility in the crude oil market can be deleterious to the business and manufacturing sectors across these economies. This is simply because volatility hampers the expectations of businesses and industries, which disrupts the production and distribution chains of consumables and affects the international supply chain of goods and services(Chen and Hsu, 2012; Sun et al., 2022b).

In the short term, the consequences of a volatile crude oil market are fatal, given that firms and businesses fail to cope with sudden price changes. Consequently, firms are forced to switch to alternative energy options (i.e., coal), which have far-reaching negative environmental footprints (i.e., in the long run, many nations must pay a high environmental cost in the form of global warming). Firms that cannot switch to crude oil substitutes are forced to shut down, leading to joblessness, recessions, and other economic policy uncertainties. Energy market volatility induced by inaccurate forecasting can lead to political instability and unexpected regime changes(Herrera et al., 2019; Oguz and Peker, 2023; Vadlamannati and de Soysa, 2020). Inaccurate forecasting of crude oil prices can also lead to a crackdown on a firm's structure, as it fails to reach critical production and management-related decisions, which are vital for the interim risk management of the firm(Balcilar et al., 2018; Cunado and Pérez de Gracia, 2003; Hamilton, 1996; Huntington, 1998; Le et al., 2021; Tian et al., 2021; Zhao et al., 2016). A consistently risky and volatile energy market encourages venture capitalists and other potential investors to look for alternative, less-volatile markets for investment, which can have pernicious consequences on energy stock prices(Batten et al., 2017; Coskun and Taspinar, 2022; Lin and Wesseh, 2013; Sun et al., 2019).

Between 1960 and 2000, the world witnessed four major oil price shocks (i.e., in 1973, OPEC introduced its first oil embargo that pushed the crude oil price from US$ 3.4 to US$ 13.4; the 1979 regime changes in Iran disrupted the supply of crude oil; the 1990s Iraqi invasion of Kuwait escalated the global crude oil price significantly; and finally 1999's oil price shock inflated the oil price from US$ 12 to US$ 24/barrel). The effects of these shocks have been widely assessed to see how these shocks affect the aggregate demand and supply of an economy, or more specifically, how macroeconomic parameters respond as an oil-based structural change materializes in an economy. (Hamilton, 1996; Huntington, 1998).A series of the negative consequences of crude oil price volatility has raised concerns among researchers, energy economists, market agents, and energy stakeholders. As such, ensuring a correct and reliable forecast of crude oil prices is paramount and time consuming. Moreover, volatility is an integral part of the risk management domain (Lv and Shan, 2013), which helps figure out the total value of the anticipated risk. Accurate forecasting has critical theoretical and applied implications, making it an integral tool for risk management, especially in the context of commodity markets where negative shocks to prices can have dire consequences (Ben Ameur et al., 2023) In such cases, machine learning and deep learning can be instrumental in minimizing risks in commodity markets by forecasting market conditions by accounting for future variations in prices. This is where technology works like magic by feeding big data into the deep learning system and generating reliable outcomes about what to expect from a commodity market in the oncoming periods. The use of technology in the form of machine and deep learning has appeared because of the limitations of other conventional data mining and processing approaches that do not predominantly rely on advanced machine learning algorithms. In this study, we rely on advanced machine algorithms that are powered by advanced technologies to unravel forecasting issues in commodity markets, especially the crude oil market, which makes technology the central focal point of this study.

The primary issue with the forecasting accuracy of the crude oil market is that it is highly complex owing to its complex mechanism, inherent structural dynamics, number of agents involved in the market, ineffectiveness of the efficient market hypothesis in the crude oil market, (Liu and Lee, 2018) and inelastic demand and supply dynamics. To reap the most positive externalities of the crude oil market and ensure holistic development for all participating agents, it is inevitable that we undertake a reliable forecasting approach that has rigor in it and helps with effective risk management and critical decision-making in the energy industry. The streaming literature in this context delineates that the topic of how we can obtain a more robust market forecasting in the energy market has received scant attention compared to the forecasting of other macroeconomic time-series (i.e., interest rate, unemployment rate) (Dutta et al., 2020; Jana and Ghosh, 2022; Nomikos and Pouliasis, 2011; Rahman, 2016). The existing literature on energy price forecasting has predominantly deployed the conventional econometric approaches, which have limitations in capturing the dynamic and structural demand and supply shocks of the crude oil market. As such, the accuracy of these forecasts has been highly debated. For instance, most earlier studies have deployed the Autoregressive Integrated Moving Average (ARIMA), Autoregressive Conditional Heteroscedasticity (ARCH), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models (Cheong, 2009; Girish, 2016; Salisu and Fasanya, 2013; Wei et al., 2010). However, the fundamental loopholes of these models are that these models assume stringent linearity assumptions and that the prices must be either linear or non-linear. Moreover, in the presence of structural breaks and evidence of robust regime change, these traditional models perform poorly (Hamilton and Susmel, 1994; Lamoureux and Lastrapes, 1990; Lin and Wesseh, 2013). Furthermore, the traditional GARCH models employ a higher rank of persistence to the conditional volatility, showing that the lagged conditional variance loses its power as it appears in the current period.

To overcome the limitations of the traditional forecasting models, some studies have applied the Markov-switching approach, where the authors claim that the Markov-switching approach has superior power in forecasting energy prices compared to the general GARCH models, especially in the event of out-of-sample forecasting(Fong and See, 2002; Lin and Wesseh, 2013; Smith, 2012). In the domain of econometric modeling frameworks to forecast the behavior of a time-series, there is no single best model that can surpass other models in terms of cumulative benefits. About the superiority of one model over the other, the observations are equivocal. Hence, the quest to find better models with higher forecasting accuracy continues. As a result, market forecasting using machine learning and advanced data-driven technologies such as Artificial Intelligence (AI) is becoming more popular among scholars and energy stakeholders worldwide. Under the concept of machine learning, machines learn how to perform the most challenging tasks using simulated data and high-end artificial intelligence (AI), which helps capture multiple algorithms. The most significant advantage of machine learning and AI is that they can capture the camouflaged non-linear trend within a time series(Herrera et al., 2019), which helps distinguish different random elements of the model that conventional econometric approaches do not adhere to.

Conventional techniques have limited ability to tackle big data, making these approaches less appealing for data management firms that deal with market forecasting and the behavior of different market participants. Big data play a pivotal role in organizing market orders and preventing market failure in the long term (Zheng et al., 2022). The rationality of using big data in forecasting the commodity markets (i.e., fossil fuel markets) has been articulated by earlier studies (Alfeus and Nikitopoulos, 2022; Ben Ameur et al., 2023; Boubaker et al., 2022), supporting the view that big data wind downs the risk of market failure by stemming market imperfections that can materialize through inaccurate forecasting. Working with big data, especially for commodity markets, requires a complementary relationship between the data and the data management system to avoid any failure in the data processing and analysis processes. When a system is fed with more useful data than usual, it becomes data rich and creates robust estimates related to market forecasting. Harnessing big data and managing it through advanced deep learning processes are fundamental, especially for commodity markets. This is because macroeconomic factors, such as aggregate demand and supply, in a commodity market may show unpredictable patterns due to exogenous economic shocks, creating an uninvited platform for market failure. In such cases, advanced machine learning and deep learning processes can be extremely helpful in preventing markets from premature crackdown by providing enough reliable information to all participating agents about what to expect in the future. This is where the newly emerged advanced deep learning methods perform better than the conventional approaches, such as the Artificial Neural Networks (ANN). Artificial Neural Networks (ANN) are a commonly used machine-learning approach to forecast energy prices; however, this approach is not beyond loopholes. The first major limitation is that this approach leads to model "overfitting" (i.e., an inaccurate model that should not fit the data well still fits), and the second drawback is the "curse of dimensionality" (i.e., dimensions additionally incorporated into the neural network make the inherent model more complex). In this milieu, studies suggest deploying hybrid models (i.e., a cross between a supervised ANN model and other unsupervised ARIMA or similar models) such as the Hybrid Neuro-Fuzzy Approach, Fuzzy Neural Networks, Support Vector Machines (SVMs), and Radial Basis Function (RBF) networks.

Based on the points mentioned above, we can postulate the following views:1) In light of the inherent dynamics of conventional econometric approaches (i.e., general GARCH, ARCH, and ARIMA), these models are incompetent for forecasting the inherently complex crude oil market, especially in the context of out-of-sample forecasting. 2) There is no model that can be labeled as "superior" compared to other "less efficient" models that can capture forecasting accurately and effectively. This is because many exogenous factors can affect the demand and supply sides of the crude oil market. 3) The incapability of existing conventional techniques has created a space for better market-oriented research, where the quest now is to find a workhorse forecasting approach built on computer-based data-driven algorithms. With access to information, informational globalization, and high-end AI, computer-simulated data-driven forecasting models appear to be better substitutes than traditional approaches. Hybrid models generate even more promising results for forecasting energy market volatility. Overall, by considering both the empirical and theoretical underpinnings of streaming literature, this study makes the following contributions to the literature. First, crude oil is the most consumed natural resource in the world, and there are many cases where territories predominantly rely on this input for electricity generation, residential usage, industrial manufacturing, etc. Due to its complex nature, volatility in the crude oil market can lead to economic deprivation, sluggish economic growth, policy uncertainty, political unrest, business downfalls, and capital market crackdowns. Based on a holistic assessment, the consequences of this volatility in the short run can be devastating, as agents of the crude oil market have less time to make informed decisions. Thus, we need marketing information that is trustworthy and well structured. Because conventional econometric approaches are not capable of propagating marketing information that is trustworthy and well-structured, we deploy artificial intelligence (AI) and machine learning-based methods to forecast crude oil price volatility. To carry out this objective, we employed hybrid GARCH-LSTM–LSTM and GARCH-GRU approaches with hyperparameter tuning. These hybrid models can capture the effects of both regulated and unregulated AI-based models. By deploying these hybrid models, we forecast crude oil price volatility by considering heterogeneous frequencies (1 min, 5 min, 30 min, 60 min, daily, weekly, and monthly). To our knowledge, this is the first study to capture the volatility of crude oil prices by harnessing both marginalized and generous time frequencies. Second, this study applies the concept of hyperparameter tuning or hyperparameter optimization to reap the most from machine learning and AI-based models. This process aids in obtaining a competitive model by boosting its performance, which is more capable of forecasting the volatility of the crude oil market with fewer errors. The application of hyperparameter tuning, or hyperparameter optimization, requires high-end computing powers and skills, which is why earlier studies disregarded this aspect while forecasting one of the inherently complex markets of the world. Third, this study adds to the energy price forecasting literature by assessing why policymakers should rethink new AI-based approaches to crude oil price forecasting. Through this study, we empirically scrutinize the rationality of using hybrid AI-based models (GARCH-LSTM, GARCH-GRU), which we believe will help control opposing market forces that lead to a collapse of the crude oil market in the short and long run.

Based on the above discussion, the study carries out the following research questions (RQ) and research objectives (RB).

RQ1. How can the forecasting accuracy of crude oil prices be improved to mitigate the impact of market volatility on global economic stability?

RQ. What role can advanced machine learning algorithms play in enhancing the reliability and precision of crude oil price forecasts?

RB1. To investigate the effectiveness of conventional econometric and modern machine learning techniques in forecasting crude oil prices, identifying limitations and areas for improvement.

RB2. To develop and test advanced hybrid forecasting models that integrate machine learning algorithms with traditional econometric methods to enhance predictive accuracy and reliability in the face of market volatility.

This study introduces innovative hybrid forecasting models that combine GARCH with LSTM and GRU neural networks to tackle the inherent complexities of crude oil price dynamics. It advances the literature on energy price forecasting by demonstrating the superiority of hybrid models over traditional econometric models in handling nonlinear dependencies and structural breaks. The research provides empirical evidence supporting the adoption of AI-driven approaches in economic forecasting, offering significant implications for policymakers and stakeholders in the energy sector.

The rest of the paper is organized as follows: Segment 2 delineates the findings of the existing studies in the domain of gas price forecasting and volatility, Segment 3 discusses the empirical approaches adopted to carry out this study, Segment 4 summarizes the results and discusses them, and Segment 5 wraps this empirical endeavor with a conclusion and policy recommendations.

2. Review of literature

Data science, in tandem with advancements in digital and satellite technology, has expanded the scope of possible AI applications in fields such as sentiment analysis and stock and commodity market predictions (Ferreira et al., 2021). As an increasing number of datasets have become available, it is becoming more common to use machine learning (ML) methods for economic information. In asset pricing, the use of linear regression approaches is constrained by the increasing number of noisy and related explanatory economic factors (Cochrane, 2011; Berger, 2023). To fully grasp investor emotions and the tweet method for stock price prediction, machine learning is always being created and enhanced. Today's investors research and make decisions based on information gleaned from internet sources such as social media and news websites (Koratamaddi et al., 2021). Considering oil's central role in modern economies, price swings in the commodity's crude form may have far-reaching consequences for national and international economies. As a globally traded commodity, crude oil prices are not affected by the availability of crude oil or the status of the economy, but rather by the market's expectations of supply and demand patterns (Gao et al., 2022). Changes in the price of crude oil have a significant effect on several other macroeconomic indices such as inflation, economic growth, currency exchange rates, and the overall trade balance between countries. From 0.5 % to 4.5 %, crude oil affects global GDP. The fluctuation in oil prices, which reflects the interplay of multiple political and military upheavals, economic shifts, conflicts, and wars throughout the world, is the single most influential factor in the correlation between crude oil and GDP. Consequently, forecasting the volatility of crude oil prices on the global market is of immense scientific and practical significance (Shehabi, 2022). Alshater et al. (2022) note in their article that the Russian invasion of Ukraine has worsened energy producers' unpredictability. Under such ambiguous conditions, an Early Warning System (EWS) might aid in efficiently stabilizing market movements. This will strengthen the capacity to reduce operational expenses and set the stage for swift economic recovery. (Chai et al., 2022) conducted a study where they constructed a time-varying parameter vector autoregression model (TVP-VAR). Through a comparative analysis between pre-COVID-19 and during-COVID-19 sample groups, their research confirmed the presence of non-linear and dynamically changing relationships among the three variables. These variables show nonlinear connections, specifically between green bonds, clean energy, and stock prices.

There is a concern that the reviewing process in the social sciences (both for empirical and review publications) has the researcher's own unacknowledged biases) (Tranfield et al., 2003; Snyder, 2019). With this constraint in mind, we took a cautious approach to the literature review for this study, conducting an in-depth, objective, and rigorous evaluation of the literature about the research subjects using a well-defined method (Rowe, 2014). To fill in the research gaps and tackle the most pressing research topics, we conducted a literature review by first mapping and then analyzing the state of current intellectual knowledge (Tranfield et al., 2003). The spillover connectedness network is highly responsive to prevailing market conditions (Khalfaoui et al., 2022).

The literature reveals the dominant research themes in a certain time, which correspond to the prevailing trends in our field of study. Volatility forecasting of crude oil and using artificial intelligence and machine learning to predict the volatility index, which is the central theme of our research, has been extensively studied over a long period of time starting in 1969, with most research taking place between 2017 and 2022. In recent years, the use of artificial intelligence (AI) and machine learning has increased rapidly in several areas, including banking, healthcare, and marketing. AI is revolutionizing company operations and can alter the nature of labor. Hence, AI and ML are used to forecast commodity market prices, with crude oil being one of the most prominent commodities. In addition, financial markets experienced global and digital recollection, focusing particularly on stock market volatility, behavioral finance, sentiment analysis, and digitally advanced financial assets. (Zhao, 2023) has implied that the global policy uncertainty index (GEPU) index on MIDAS-type models can be used to predict the volatility of oil futures. These results show that GEPU can accurately estimate the volatility of oil futures. In addition, the MIDAS model with regime switching and GEPU can enhance the accuracy of model forecasts.

Machine learning and artificial intelligence (AI) are among the most powerful digital tools that have contributed to virtually all academic disciplines. According to Christensen et al. (2022), machine learning can deal with the usually non-linear structure regulating financial markets and extract significant information about future volatility in a data-rich environment with expanding feature space, but linear regression techniques fail. In addition, machine learning (ML) algorithms dramatically increase prediction because they can analyze sentimental values, tweets, and many other variables to forecast the best outcomes. According to Gong and Xu (2022), geopolitical risk, particularly geopolitical act risk, has a substantial impact on the interconnectedness of commodities markets. Moreover, the implications of the various commodity markets on net spillovers are distinct. Geopolitical risk has favorable impacts on the net spillover of energy, agriculture, and livestock markets but negative effects on precious metal and industrial metal markets. During the Russia-Ukraine conflict, the overall volatility spillover increased from 35 % to 85 %, surpassing the amount seen during the epidemic. Wang et al. (2022) alter the function of commodities in return and volatility spillover systems.

2.1. Strategic map

A thematic map is a graphical depiction of the framework's major principles. In the context of a strategic diagram, subjects are categorized based on their centrality and density (Cobo et al., 2015). Density evaluates how effectively diverse subjects are interconnected, while centrality evaluates how well diverse topics are interconnected (Aparicio et al., 2019). The motor theme is highly central and dense in the upper right quadrant, whereas the peripheral theme is similarly dense, but less so in the upper-left quadrant. Low centrality and density are seen in the bottom-left quadrant, where the emerging or fading topic is situated, and in the lower-right quadrant, where the transversal, generic, or fundamental subject is situated.

Fig. 1 depicts a strategic map with eight themes, divided into four quadrants. Volatility forecasting, oil price, crude oil market, spillover effect, exchange rate volatility, stock market, monetary policy, crisis. Internal linkages are weaker in this quadrant, showing that more study is needed in the future to thoroughly assess the sub-themes, even though their ties to other themes have strengthened over time (Rodríguez-Soler et al., 2020). Furthermore, the size of the circles suggests that these issues have been carefully investigated; however, further academic research is necessary.

According to the study, the top two in terms of the size of significance are "volatility forecasting" and "Crude oil market speculation." For the most frequent occurrences, the phrase "volatility forecasting" contains terms such as global economic uncertainty, digital revolution, artificial intelligence and machine learning, forecasting evaluation, geo-political risk, crude oil futures, and crude oil markets (Al-Fattah, 2019; Chen et al., 2020; Mei et al., 2020; Dai et al., 2022; Kiddle, 2022; Lu et al., 2022; Zhang et al., 2023). Volatility is a crucial factor in deciding the prices of derivatives, mitigating risks in investment portfolios, and implementing hedging tactics in the financial markets. Therefore, it is crucial to accurately assess the volatility of essential natural resources such as crude oil, gold, copper, natural gas, and silver (Srivastava et al., 2023). Forecasting volatility has become important in managing uncertainty and oil shocks. Therefore, AI and ML are important tools for predicting price volatility in the oil market. Zhang et al. (2023) employed machine learning from both variable selection (VS) and common factor (i.e., dimension reduction) viewpoints. Their findings show that the supervised PCA regression model can correctly forecast in-sample and out-of-sample oil market volatility. Moreover, the suggested models can provide predictive advantages from both statistical and economic viewpoints. Zhang et al. (2022) compared the general global economic policy uncertainty (GEPU) indices to the aligned (GEPU) index using a modified machine learning approach. Out-of-sample volatility in the crude oil market can be predicted with high precision using the aligned GEPU index. Compared to generic GEPU indices that

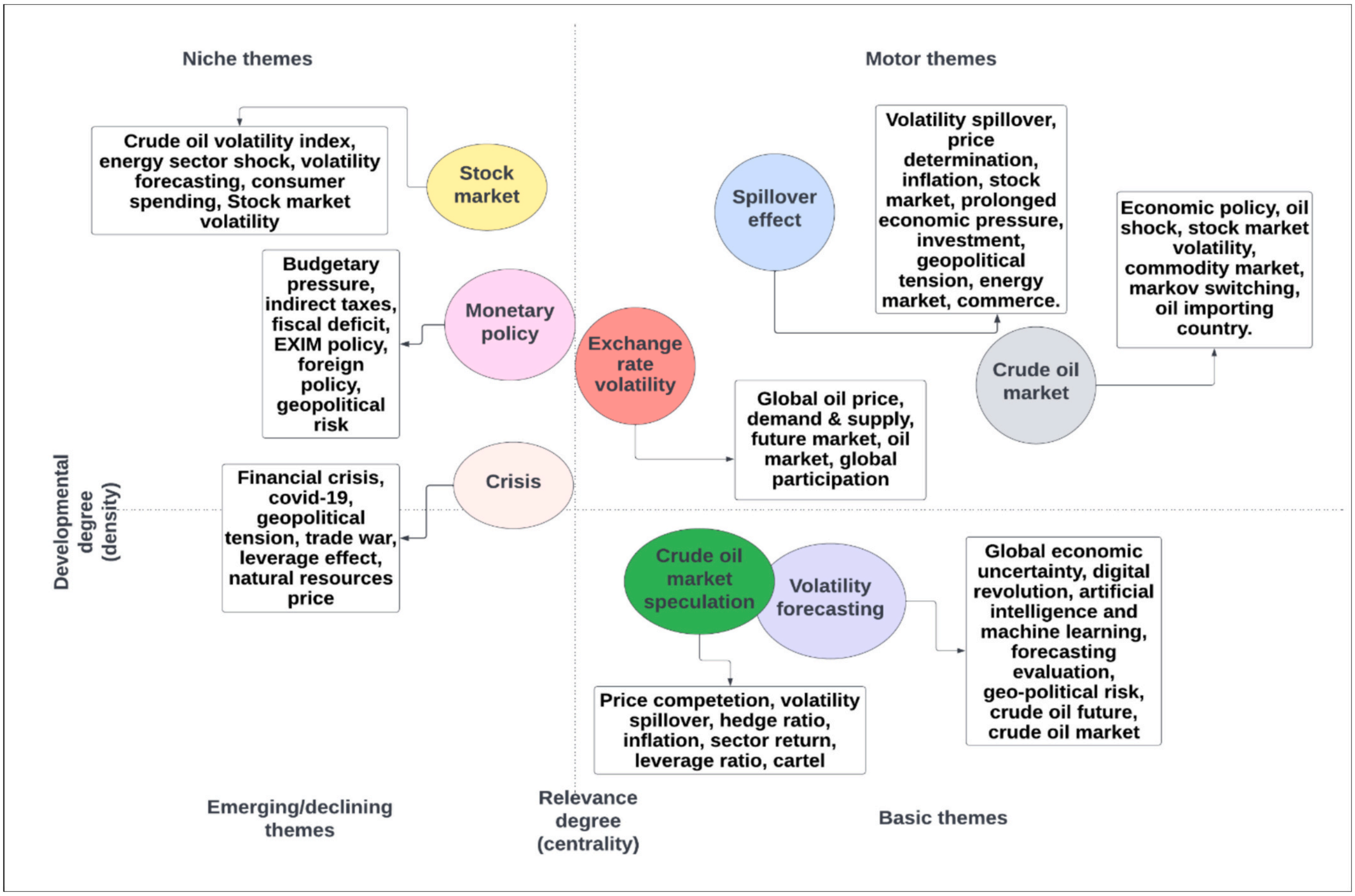

Fig. 1. Strategic map.

do not integrate supervised learning, well-known economic characteristics, and other prominent uncertainty indicators, the aligned GEPU index is highly potent and may provide either preponderant or complementary information. Therefore, incorporating ML and AI into the forecasting model will consider every angle and dimension of the data and build the solution considering that into perspective. Using the impact of jumps, leverage effect, and regime switching as the independent variables, Lu et al. (2022) forecast oil futures realized range-based volatility (RRV) in the mixed data sampling (MIDAS) framework. The results show that, even in the face of excessive volatility, the most effective RRV forecasting of oil futures requires the simultaneous combination of jumps, the leverage effect, and regime flipping. Most importantly, regime switching can boost the prediction accuracy for longer forecasting horizons, such as weekly, bimonthly, and monthly. Also, Crude oil has a strong connection to national strategy and global politics (Jiao et al., 2023). Geopolitical risks, unlike traditional risks found in financial markets such as liquidity risk, market risk, credit risk, and operational risk, are easily noticeable in their influence on financial markets (Zhang et al., 2023). These risks also have intricate effects on oil prices. Moreover, crude oil, being a significant contributor to the energy sector, is also a crucial underlying asset for investing in the commodities market. This grants it the twin attributes of being both a tangible and monetary underlying asset (Jiao et al., 2023; Zhao, 2023).

The second most popular themes, "Crude oil market speculation," which overlapped with volatility forecasting, included terms like Price competition, volatility spillover, Hedge ratio, Inflation, Sector return, Leverage ratio, and Cartel (Duppati and Zhu, 2016; Yang et al., 2019; Belhassine, 2020; Köse and Ünal, 2021; Berger, 2023). Crude oil, as the world's major energy source, is critical to a country's economy and social stability (Zhu et al., 2019). However, oil supply and demand, global economic fundamentals, economic collapse, catastrophic public health crises, diplomatic conflicts, and other internal and external variables can produce substantial changes in crude markets. (Cui and Feng, 2020; Zhu et al., 2021). Mensi et al. (2021) investigated the potential use of oil as a hedge or safe-haven asset against precious metals is investigated by (Mensi et al., 2021). These numbers point to oil futures as a smart move for investors looking to hedge against potential losses and diversify their holdings. The implications for portfolio hedgers in terms of deciding proper portfolio allocations, managing risks, and anticipating movements in the equity and commodities markets are also substantial. Ito (2012) and Köse and Ünal (2021) discussed the consequences of oil price volatility on inflation and GDP are discussed by Ito, 2012; Köse and Ünal (2021). In the long run, a 1 % change in oil prices results in 0.44 a point growth in real GDP. Similarly, we show that rising oil prices not only have negative and positive effects on inflation and economic growth in the near term (four quarters) but also result in an appreciation of the real effective exchange rate. ML and AL may be applied sector-by-sector to forecast the returns and volatility of a particular sector, as stock returns are sensitive to oil price fluctuations and leverage effects. The volatility of volatility decreased quickly prior to the war. However, it exhibits a more prolonged decline in the S&P GSCI Natural Gas Index (Chen et al., 2023).

Using the generic VAR-GARCH method, El Hedi Arouri et al. (2011) examined the amount of volatility transmission between the oil and stock markets in Europe and the United States at the sectoral level. They argue that oil prices have a significant impact on the volatility of equities in the same industry. The oil market's impact on the stock market tends to be unidirectional in Europe but two-way in the United States. Finally, our historical testing shows that hedging and diversification are improved when using estimated cross-market volatility spillovers from

VAR-GARCH models rather than more standard multivariate volatility models, such as Bollerslev's CCC-GARCH, Engle and Kroner's diagonal BEKK-GARCH, and Engle's DCC-GARCH. Carbon emissions (RCE) and crude oil (RCO) demonstrate a notable and significant correlation with both Shanghai and European stock exchanges (Yadav et al., 2023).

The third most important theme is the crude oil market, which belongs to the motor theme and includes economic policy, oil shock, stock market volatility, commodity markets, Markov switching, and oil-importing countries (Ma et al., 2019; Lu et al., 2021; Salisu and Gupta, 2021; Kannadas and Viswanathan, 2022; Lee and Kim, 2022). In recent years, unprecedented changes in crude oil prices have been seen. For instance, on July 3, 2008, the price of a barrel of West Texas Intermediate (WTI) crude oil hit a high of $145.31 before plunging to under $40 by the year's end due to the global financial crisis. As a result of the negative demand shock caused by COVID-19, oil prices became negative in April 2020 (Liu et al., 2022). From January 2010 to May 2021, Lee et al. (2022) used a structural vector autoregression framework to analyze oil shocks, geopolitical risks, and green bond returns in China. The results show that geopolitical unpredictability and the behavior of green bonds are both significantly affected by oil-specific demand shocks. Green bond profits are reduced, and geopolitical dangers increase when oil prices suddenly rise. Furthermore, the data prove that decreases in oil prices and increases in green bond returns are associated with reductions in geopolitical risks, while the impact of geopolitical shocks differs according to the risk categories under consideration. Oil prices are negatively affected by geopolitical acts, but green bond returns are significantly boosted by geopolitical threats.
The Markov switching model, also known as regime switching in literature, is a statistical modeling technique used to analyze time-series data that show changes in behavior over time. Using a Markov-switching vector correction model (MRS-VECM), in which the amount of cointegration and correlation is conditional on the regime of the volatility indices, Li (2022) explores the dynamic interrelationships between the oil and equity volatility indices. The results show that OVX (oil volatility index) is related to the VIX (volatility index of the stock market (VIX) over the long run (VIX).

The fourth most important theme is the spillover effect, which includes volatility spillover, price determination, stock markets, prolonged economic pressure, Investment, Energy markets, and commerce (Sharifi-Renani and Mirfatah, 2012; Hameed et al., 2021; Su et al., 2021). Predicting the price of crude oil is a difficult task. To enhance this prognostication, the DFN-AI model employed by Wang et al. (2018) is a hybrid approach that combines a data fluctuation network (DFN) with various AI techniques. Traders and analysts may benefit greatly from the DFN-AI method of crude oil price forecasting because it allows them to quickly decide prices based on historical data and future projections. Crude oil is an essential energy source for both commercial and industrial applications. Volatile swings in the price of crude oil (COP) have significant effects on economic stability and national security, posing significant difficulties for international financial stability and industrial management (Huynh et al., 2020). As a "industrial blood," the impact of COP fluctuations is significant. For instance, drastic differences in worldwide COP have significantly affected the sustainability of social development and economic expansion (Chai et al., 2019; Ali Salamai, 2023). There is a direct link between the volatility of crude oil prices and that of the energy market, with fluctuations in crude oil prices having a substantial effect on the cost of producing and consuming energy products. Therefore, AI can be used to assess crude oil price volatility and forecast energy market volatility. This can be carried out by utilizing machine learning algorithms, which can evaluate vast quantities of historical data on crude oil prices and other pertinent aspects, such as global demand and geopolitical events, to discover trends and make forecasts (Fianu, 2022; Srivastava et al., 2023). (Hasan et al., 2024) proposed a novel approach to forecasting crude oil prices using blended ensemble learning by combining various machine learning methods and validate their model with Brent and WTI crude oil price data. The proposed model outperforms existing methods in forecasting errors for short- and medium-term horizons.

The fifth significant theme is "Exchange rate volatility" which includes Global oil price, demand & supply, Future market, Oil market, and Global participation (De Jesús Gutiérrez, 2018; Tong et al., 2018; Yang et al., 2021). Ghosh (2011) in her study revealed that exchange rate connections are volatile over time. An increase in the oil price return causes the Indian rupee to depreciate against the US dollar. The theoretical connection between oil prices and currency rates is reassuring for oil-importing countries. Considering the current and future rise in global oil prices, Indian refineries would need to acquire more dollars to pay for oil imports, which would drive down the value of the Indian rupee. Sun et al. (2022a, 2022b) analyzed the impact of fluctuating oil prices on China's manufacturing sector is analyzed by (Sun et al., 2022a). According to his findings, the production and consumption of crude oil-refined coke products and processed nuclear fuel products are more sensitive to changes in the price of crude oil. If the crude oil price fluctuates significantly, the public utilities sector will be the hardest. Owing to the unstable price of crude oil, investments have been cut significantly. To keep consistent investment and output across all sectors, the crude oil price must remain stable.

As a nation that imports oil, this oil price–exchange rate link is consistent with economic theory. Therefore, an increase or expected increase in international oil prices forces Indian refineries to buy additional dollars to pay for more expensive oil imports, resulting in a devaluation of Indian rupees. Sun et al. (2022a, 2022b) assessed the effects of oil price hikes and declines in China's industrial sectors. Their results showed that crude oil price fluctuations have the largest effect on the production and consumption of crude oil refined coke products and processed nuclear fuel products. The public utilities sector is the most vulnerable to variations in crude oil prices. As the price of crude oil fluctuates, investments drastically reduce. Stability in the price of crude oil is crucial for investment and production across all industries.

The sixth important theme is "Monetary policy" which includes Budgetary pressure, Indirect taxes, Fiscal deficit, EXIM policy, foreign policy, and Geopolitical risk (Chakraborty, 2022; Elhannani, 2022; Ma et al., 2022; Majumder et al., 2022). Crude oil prices can have far-reaching effects on a country's monetary policy. Because crude oil is used in the manufacturing of several products, fluctuations in its price can have a significant impact on the cost of production and, in turn, on the pricing of those products. This, in turn, may affect inflation, one of the primary targets of central banks when deciding monetary policy (Ramyar and Kianfar, 2019). Uncertainty and unpredictability may work upon the economy when crude oil prices are erratic. To better prepare central banks and policymakers for economic shifts, crude oil volatility can be predicted using AI and machine learning approaches. To decide how people feel about the current price of crude oil, natural language processing (NLP) techniques may be used to sift through news stories, social media, and other sources of information. This can be useful in pinpointing the origins of fluctuations, such as political unrest or shifts in supply and demand (Charles et al., 2022; Claus and Stella, 2022). The available research has thus far largely depended on the auto regression VAR-GARCH model to predict crude oil volatility, but the cutting-edge digital revolutionary approach of artificial intelligence has performed extremely substantially and accurately in predicting the volatility of not only crude oil but also the stock market and commodities markets. Crude oil volatility may affect a country's export–import (EXIM) policy by affecting import and export prices and trade balance (Gondaliya, 2015; Saputera et al., 2021). Therefore, the literature has also given importance to AI and machine learning, which can forecast crude oil volatility, helping policymakers predict and modify EXIM policies for global oil market fluctuations (Liang et al., 2023). According to the literature on AI and ML, AI has been employed to analyze satellite photographs of oil ship movement and storage facilities to estimate real-time crude oil supply and demand and expect volatility (Toorajipour et al., 2021). The use of AI and ML will enhance sustainability and solve issues in the supply chain.

The seventh theme is the stock market, which includes the crude oil volatility index, energy sector shocks, volatility forecasting, consumer spending, and stock market volatility (Wang, 2013; Dutta, 2017; Bashir, 2022; Raggad, 2023). (Luo and Qin, 2017) in their research mention that the evidence suggests that crude oil volatility index (OVX) shocks have substantial and negative impacts on the stock market, but the impact of realized volatility shocks is insignificant, particularly following the current financial crisis. Therefore, it is very important for regulators, policy makers, and investors to have correct and predictive knowledge about the movement of the prices of crude oil so that they act accordingly to avoid maximum damage. Market transmission has risen because of financialization and improved global market integration. It is critical to understand the types and amount of links between various financial markets (Ma et al., 2023; Rao et al., 2023). Wang (2013) examined the influence of increased oil prices on consumer spending in open and industrialized nations. The findings suggest that as oil prices rise, consumer spending decreases; however, the opposite is not true when prices fall. The true balance and smooth transition impacts of changes in oil prices affect individuals' discretionary spending habits. As the oil price increases, consumers cut back below a certain level. Customers are logically recommended to postpone purchases until future oil prices are known. Inflationary pressures have now caught up with the costs of local production factors. Therefore, it is very important for marketers and product manufacturers to have knowledge of the volatility of oil prices to frame the policy and recommend their manufacturing division on the tendency of consumer spending. According to Valecha et al. (2018), artificial intelligence algorithms may be used to forecast customer spending and buying habits, making it a wise investment for businesses to include these technologies in their product development and marketing efforts. Chatziantoniou et al. (2022) investigated the ripple effect of volatility in the G7 stock and crude oil markets from 2007 to 2021. The results show that the network's degree of interconnectedness responds strongly to shocks that have a major impact on global financial markets and considerable value across the time span of the sample. A significant portion of volatility shocks are transmitted across the network via the crude oil market. Therefore, crude oil volatility has a significant impact on the financial markets of the G7 nation.

The volatility of crude oil prices has a significant impact on the financial markets of the G7 nations, comprising Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States. This impact is multifaceted, affecting various aspects of these nations' economies. Firstly, in stock markets, fluctuations in crude oil prices can influence the performance of energy sector companies, leading to corresponding changes in stock prices and broader market indices. For countries that are net exporters of oil, such as Canada, changes in oil prices can affect currency markets, with currency values often strengthening or weakening in response to oil price movements. Central banks in these nations also closely monitor oil price changes, as they can impact inflation rates. High oil prices can contribute to inflation, leading central banks to consider raising interest rates to control inflationary pressures. Additionally, fluctuations in oil prices can affect consumer spending patterns, as higher oil prices can result in increased costs for transportation and heating, potentially reducing disposable income and impacting consumer confidence and spending. Overall, the volatility of crude oil prices is closely watched by policymakers, investors, and analysts in G7 nations due to its potential to influence financial markets, inflation, interest rates, consumer spending, and overall economic growth.

The final theme is "Crisis" which is partially in the emerging theme and niche theme as it very significant and gradually moving toward the motor theme it includes Financial crisis, Covid-19, Geopolitical tension, Trade war, Leverage effect, Natural resources price (Singh and Singh, 2017; Lee et al., 2021; Zhu et al., 2021; Liu and Chen, 2022; Lu et al., 2022; Pan et al., 2023). The Covid-19 issue is a horrific human tragedy that has thrown the world economy into disarray by altering supply and demand. This disruption has had a substantial influence on enterprises, commerce, and people's lives, causing uncertainty in supply chains, consumption, production, financial stress, and product pricing. Furthermore, the pandemic has had an impact on the stock market, oil markets, currency markets, and commodities markets (Bouazizi et al., 2024).

Yu et al. (2022) examine how changes in oil prices affect the macroeconomy during global financial crises and pandemics. Despite the prevalence of global financial crises, the available literature shows that both economic activity and oil prices have always demonstrated remarkable resilience. The current COVID-19 epidemic showed significant volatility in economic activity and oil prices during the crisis period. Furthermore, the data reveals a connection between oil prices and economic activity amid global financial crises and COVID-19. According to the authors, the global financial crisis and COVID-19 outbreak, both of which caused a spike in oil prices, had severe consequences for economic activity. Compared to earlier global financial crises, the COVID-19 outbreak had the highest overall connectivity between oil prices and economic activities, suggesting that information transmission between oil prices and economic activities occurs more quickly during this time (Jiang et al., 2020). The enormous changes in oil prices over the past few decades have made the crude oil markets unpredictable and hazardous. Chen et al. (2019) in their paper examined the direct impact of the leverage effect on risk indicators, such as VaR and CVaR, as well as assessing the presence of the leverage effect by analyzing daily spot returns in the WTI and Brent crude oil markets. The findings suggest that oil-demanding enterprises utilizing VaR for risk management are concerned about losses exceeding VaR, while minimizing the opportunity cost of capital may benefit from leverage. Alshater et al. (2022) used a proxy for energy market volatility, such as an uncertainty index in a Machine Learning (ML) model, to predict energy equity prices. Combining network models with ML to optimize risk management tactics makes ML a more potent tool for forecasting and predicting market risks, and an efficient tool for predicting natural resource prices. Papadimitriou et al. (2014) investigated the accuracy of forecasting directional changes in power prices using forecasting models based on Support Vector Machines (SVM). Their results suggest that SVM is a viable method for forecasting near-term power costs, with a projected accuracy of 76.12 % over a 200-day timeframe. Therefore, ML and AI are gifts from the digital revolution to predict prices and act proactively in challenging situations.

3. Data and methodology

3.1. Data

This study employed an empirical analysis of crude oil brand prices using various time frequencies and sample periods. The objective was to investigate the forecasting performance of different models, including GARCH (1,1), Vanilla LSTM, GARCH (1,1) LSTM, and GARCH (1,1) GRU, on crude oil price data. The data used in this study consisted of daily and intraday observations of crude oil prices. The first dataset includes daily observations from February 10, 2003, to November 14, 2023, with a total of 5021 observations. The second dataset contains 60-min interval data from January 1, 2015, to February 13, 2023, with a total of 46,264 observations. The third dataset includes 30-min interval data from October 2, 2022, to February 12, 2023, with a total of 4159 observations. Additionally, two samples with 1-min interval data were used. The first sample, referred to as Sample I, includes data from November 30, 2022, to February 12, 2023, with a total of 43,339 observations. The second sample, referred to as Sample II, includes data from January 1, 2022, to March 31, 2022, with a total of 61,272 observations.

1. Crude Oil Brent - Daily (10 Feb 2003–14 Nov 2023, 5021 observations):

- Rationale: Daily frequency is commonly used for GARCH models to capture medium to long-term volatility patterns in financial markets. The selected sample period covers a substantial timeframe, allowing for the analysis of various market conditions and trends.

- Methodological Robustness: Daily data provides a good balance between capturing volatility patterns and reducing computational complexity. The long sample period enhances the robustness of the analysis by including data from different market regimes.

2. Crude Oil Brent - 60 min (2023-02-13 to 2015-01-01, 46,264 observations):

- Rationale: 60-min frequency provides a higher granularity than daily data, capturing more short-term volatility patterns. The extended sample period allows for a detailed analysis of intraday price movements.

- Methodological Robustness: The use of 60-min data enhances the model's ability to capture short-term dynamics and intraday patterns. The long sample period ensures robustness by including data from various market conditions over several years.

3. Crude Oil Brent - 30 min (2023-02-12 to 2022-10-02, 4159 observations):

- Rationale: 30-min frequency offers even higher granularity, suitable for capturing more detailed intraday price movements and volatility patterns.

- *Methodological Robustness*: The use of 30-min data allows for a more precise analysis of short-term dynamics. Although the sample period is shorter, it still covers a significant timeframe, ensuring the inclusion of diverse market conditions.

4. Crude Oil Brent - Sample I: 1 min (2023-02-12 to 2022-11-30, 43,339 observations):

- Rationale: 1-min frequency provides the highest granularity, capturing very detailed intraday price movements and volatility patterns.

- *Methodological Robustness*: The use of 1-min data allows for a highly detailed analysis of short-term dynamics, providing valuable insights into intraday market behavior. The extended sample period ensures robustness by including a large number of observations over several months.

5. Crude Oil Brent - Sample II: 1 min (2022-03-31 to 2022-01-0, 61,272 observations):

- Rationale: Similar to Sample I, 1-min frequency is used to capture detailed intraday price movements and volatility patterns.

- Methodological Robustness: The use of 1-min data in Sample II provides additional insights into intraday market behavior, complementing the analysis from Sample I. The long sample period ensures robustness by including data from various market conditions over several months. In summary, the selection of specific time frequencies and sample periods for each model is based on the desired level of granularity and the need to capture different aspects of volatility patterns in crude oil prices. Each dataset provides valuable insights into market dynamics at different levels of detail, contributing to the overall methodological robustness of the research.

3.2. Methodology

The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is a popular time-series model widely used in financial modeling and forecasting. It was first introduced by Bollerslev in 1986 and is a variation of the autoregressive conditional heteroscedasticity (ARCH) model. The GARCH (1,1) model is one of the most common specifications in the GARCH family of models.

The GARCH (1,1) model is specified as follows:

$$yt = \mu t + \varepsilon t \tag{1}$$

where yt is the observed time series; μt is the conditional mean of the series at time *t,* and εt is the error term. The error term εt is assumed to be independently and identically distributed (i.i.d.), with zero mean and unit variance.

The conditional variance of the series at time t is given by:

$$\sigma t^2 = \omega + \alpha 1 \varepsilon t - 1^2 + \beta 1 \sigma t - 1^2 \tag{2}$$

where σt^2 is the conditional variance of the series at time *t,* ω is a constant term, $\alpha 1$ and $\beta 1$ are the coefficients of the model, and $\varepsilon t - 1^2$ and $\sigma t - 1^2$are the squared residuals and the squared conditional variance at time *t-1* respectively.

The GARCH (1,1) model assumes that the error terms are serially uncorrelated, but their variance is time varying and dependent on the past values of the series.

Estimate a GARCH (1,1) model on the training set:

$$\sigma_t^2 = \omega + \alpha^* \varepsilon_{(t-1)}^2 + \beta^* \sigma_{(t-1)}^2 \tag{3}$$

where σ_t^2 is the conditional variance of crude oil prices at time t, ω is a constant term, α is the weight of the past error term $\varepsilon_{(t-1)}^2$, and β is the weight of the past conditional variance $\sigma_{(t-1)}^2$.

LSTM Model Training:

Train an LSTM model on the training set:

$$h_t = f\left(W^* x_t + U^* h_{(t-1)} + b\right) y_t = g(V^* h_t + c) \tag{4}$$

where h_t is the hidden state at time t; f and g are activation functions; W, U, and V are weight matrices; b and c are bias vectors; x_t is the input vector at time t (which contains the GARCH(1,1) residuals and possibly other variables); and y_t is the output at time t (which represents the predicted volatility).

Hybrid Model Prediction:

The trained GARCH (1,1) and LSTM models were used to make predictions on the testing set as follows:

$$\sigma_t^2 = \omega + \alpha^* \varepsilon_{(t-1)}^2 + \beta^* \sigma_{(t-1)}^2 + \lambda^* y_t \tag{5}$$

where λ is the weight of the LSTM output y_t, which captures additional information from the non-linear relationships captured by the LSTM model.

3.3. GRU

3.3.1. GRU model training

Train a GRU model on the training set:

$$h_t = (1 - z_t)^* h_{(t-1)} + z_t^* tanh\left(W^* x_t + U^* r_t^* h_{(t-1)} + b\right) y_t = g(V^* h_t + c) \tag{6}$$

where h_t is the hidden state at time t; z_t is the update gate; r_t is the reset gate; f and g are activation functions; W, U, and V are weight matrices; b and c are bias vectors; x_t is the input vector at time t (which contains the GARCH(1,1) residuals and possibly other variables); and y_t is the output at time t (which represents the predicted volatility).

Hybrid Model Prediction:

We used the trained GARCH (1,1) and GRU models to make predictions on the testing set.

$$\sigma_t^2 = \omega + \alpha^* \varepsilon_{(t-1)}^2 + \beta^* \sigma_{(t-1)}^2 + \lambda^* y_t \tag{7}$$

where λ is the weight of the GRU output y_t, which captures additional information from the non-linear relationships captured by the GRU model.

Model	Characteristics	Strengths	Limitations
GARCH (1,1)	Time series model for volatility clustering	Simple and effective for volatility patterns	Assumes linear dependence on past squared obs.
LSTM	RNN architecture for sequential data modeling	Captures long-term dependencies, avoids vanishing gradient problem	Computationally expensive, may overfit
GARCH (1,1) LSTM	Combines GARCH (1,1) with LSTM architecture	Captures volatility patterns and long-term dependencies	Complexity and computational requirements
GARCH (1,1) GRU	Combines GARCH (1,1) with GRU architecture	Balances complexity and effectiveness	May not capture complex sequential patterns as effectively as LSTM

4. Empirical results

Table 1 shows the characteristics of the crude oil brand prices. The variable of focus is crude oil Brent, and the frequency of recording the prices can be daily, every 60 min, every 30 min, or every minute. The sample period for which crude oil Brent prices were recorded was mentioned for each frequency. For example, the Brent crude oil prices were recorded daily between February 10, 2023, and November 14, 2003, which constitutes a period of 5021 observations. Similarly, the table shows that crude oil Brent prices were recorded every 60 min for an extended period starting from February 13, 2023, to January 1, 2015, with a total of 46,264 observations. The third row shows that Brent crude oil prices were recorded every 30 min between February 12, 2023, and October 2, 2022, resulting in 4159 observations. Furthermore, the table indicates that crude oil Brent prices were recorded for two different 1-min samples: Sample I between February 12, 2023, and November 30, 2022, comprising 43,339 observations, and Sample II between March 31, 2022, and January 1, 2022, constituting 61,272 observations. In summary, the table presents the frequency, sample period, and number of observations for Brent crude oil prices, providing insight into how frequently the prices were recorded and over what time periods.

Fig. 2 offers a detailed representation of daily crude oil prices from 2000 to 2023. In the initial phase, spanning 2000 to 2008, there was an evident surge in prices, reflecting the era of profound global economic expansion. This growth reached a pinnacle in 2008, the point at which the global financial crisis struck, as demarcated by the first vertical dashed line. The aftermath of this crisis saw a rapid plummet in oil prices, attributable to the severe contraction in worldwide demand. As the world gradually recuperates from the financial downturn, oil prices oscillate, but never regain their zenith before the crisis. This dynamic underwent another shift between 2014 and 2016, marked by the second vertical dashed line indicating the surge in US shale oil production. The proliferation of shale oil effectively boosts global oil supply and exerts downward pressure on prices. This new equilibrium persists until around 2020, when the graph delineates another crucial juncture: the advent of the COVID-19 pandemic. As nations grapple with the pandemic, imposing lockdowns, and restricting travel, a precipitous decline in oil demand ensues, leading to a sharp drop in prices. However, resilience is evident as prices rebound after this phase, reflecting the world's efforts to revitalize economies. However, by 2023, a discernible decline signals potential challenges or market adjustments. This chronicle of oil prices, meticulously plotted over these years, underscores the volatility intrinsic to the oil market and its susceptibility to global socioeconomic and geopolitical events.

Table 1
Variable, frequency, sample duration and observations.

Variable	Frequency	Sample Period	Observation
Crude Oil Brent	Daily	10 Feb 2023–14 Nov 2003	5021
Crude Oil Brent	60 min	2023-02-13, 2015-01-01	46,264
Crude Oil Brent	30 min	2023-02-12, 2022-10-02	4159
Crude Oil Brent	Sample I: 1 min	2023-02-12, 2022-11-30	43,339
Crude Oil Brent	Sample II: 1 min	2022-03-31, 2022-01-0	61,272

Fig. 3 provides an insightful depiction of the return characteristics of the daily crude oil prices from 2000 to 2023. Beginning with the top-left panel, the time series plot of returns conspicuously oscillates around zero, mostly confined within a range of −10 to 10, but exhibits notable spikes around 2020, which alludes to significant positive returns potentially triggered by specific economic or geopolitical events. Adjacently, the top-right panel offers a histogram coupled with a density plot underlining the distribution of returns. Its bell-shaped curve, which is symmetrically centered around zero, suggests that most returns are closely packed near the mean. Delving into the logarithmic aspect, the bottom left panel shows log returns over time. Similar to raw returns, log returns are predominantly clustered around zero, albeit with occasional extreme values. The bottom-right panel reinforces this observation by illustrating the distribution of log returns. Here, a pronounced peak at the center reflects the preponderance of values near zero, whereas tapering tails capture the infrequency of extreme log returns. Collectively, this figure presents a vivid picture of the volatility and distribution dynamics of crude oil prices during the observed period.

Fig. 4 depicts the Realized Volatility Using Different Interval Windows over a span of two decades from 2004 to 2024. Realized volatility is a measure of how much the price of an asset, such as a stock, varies over a specific timeframe. The graph illustrates this volatility using different interval windows:1-day, 7-day, 14-day, and 21-day intervals. The y-axis represents the magnitude of the realized volatility, ranging from 0.00 0.16. The x-axis represents the timeline starting in 2004 and ending in 2024. From the figure, it is apparent that the 1-day interval realized volatility, represented by the blue line, shows the most fluctuations. This finding is consistent with the idea that shorter intervals tend to capture more granular day-to-day price changes, leading to higher volatility values. The 7-day (in orange), 14-day (in green), and 21-day (in red) interval volatilities are smoother in comparison, as they average the daily fluctuations over longer periods. However, they also capture broader trends and significant events that affect volatility over longer durations. One particularly notable observation is the sharp spike in volatility across all interval windows around the year 2020. This suggests that a significant market event or external factor led to heightened uncertainty and price fluctuations during that period. The volatility seems to have subsided somewhat post-2020 but remains relatively elevated compared to the pre-2020 era.

Fig. 5 shows the distribution of the daily volatility categorized by month using a combination of box-and-whisker plots and individual data point overlays. From January to December, the visualization highlighted several aspects of the data. The median daily volatility remains notably consistent across all months, slightly surpassing the 0.04 threshold. However, the interquartile range (IQR), representing the middle 50 % of the data, revealed a distinct variability. Months, such as January, February, March, and December, demonstrate a broader IQR, suggesting heightened volatility variability during these periods. Concurrently, outlier data points, representing exceptionally high- or low-volatility days, are evident in nearly every month, with months such as March, April, and December manifesting more pronounced outliers. This distribution may have intimate potential seasonal volatility patterns, with specific months displaying a predisposition to larger market fluctuations. This figure serves as a valuable tool for discerning monthly patterns and anomalies in daily volatility, which is crucial for understanding the underlying market dynamics.

Table 2 presents the forecasting performance of the four models for daily observations of Crude Oil Brent prices: GARCH (1,1), LSTM, GARCH (1,1) LSTM, and GARCH (1,1) GRU. Performance was evaluated using three widely used error metrics: Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Squared Error

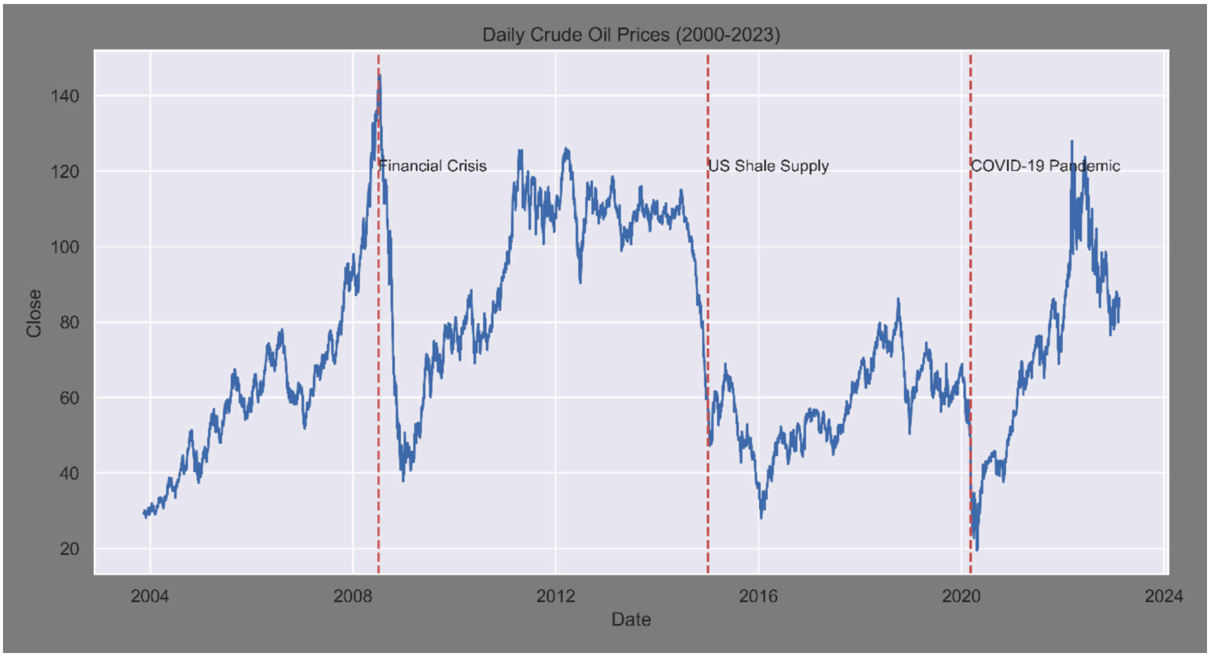

Fig. 2. Daily crude oil prices.

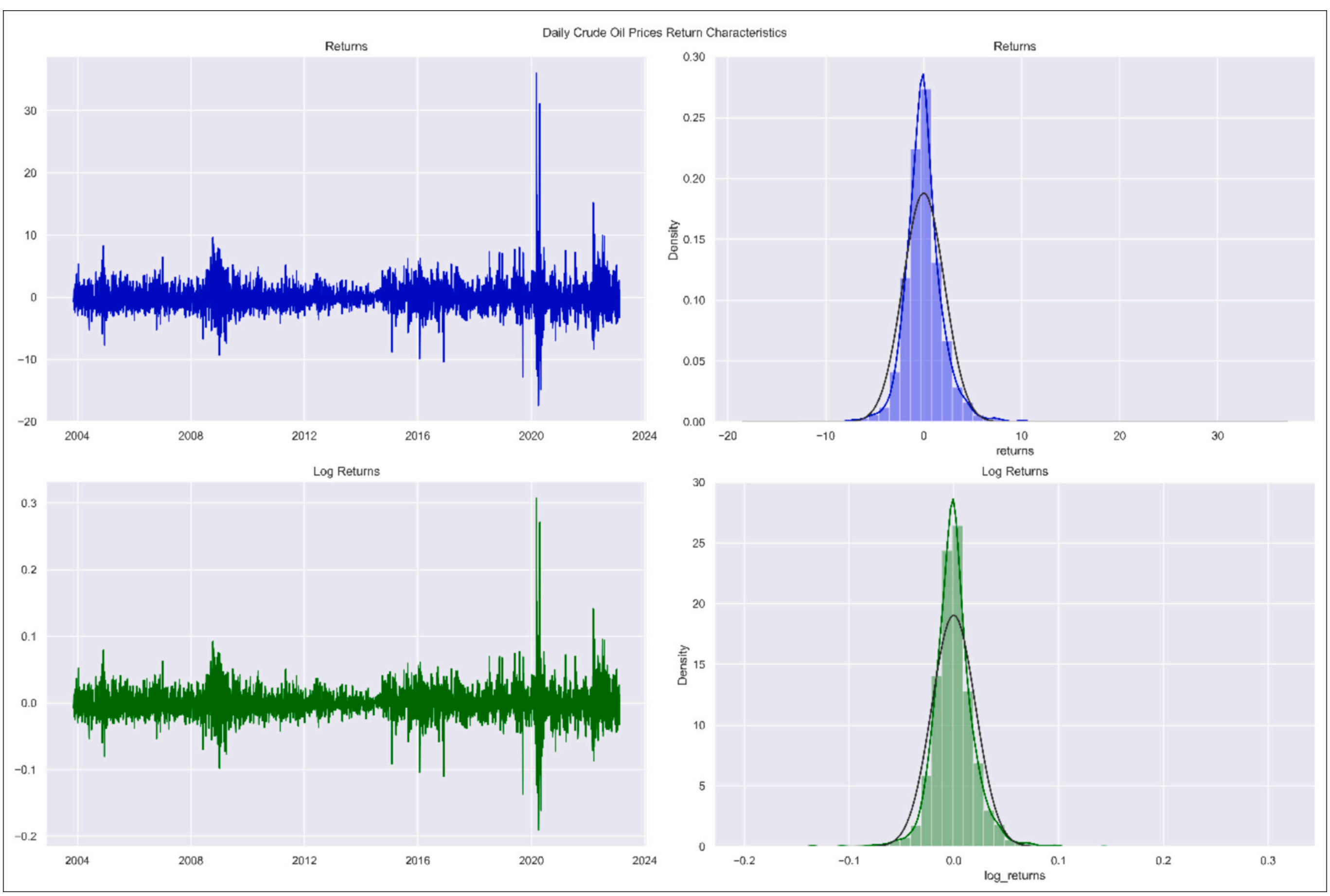

Fig. 3. Daily crude oil returns characteristics.

(MSE). GARCH (1,1), this traditional time-series model yields an RMSE of 2.8896, an MAPE of 277.82, and an MSE of 8.34844. These relatively high error values indicate that the GARCH (1,1) model's forecasting performance on daily crude oil data is not as accurate as that of the other models. The deep learning model, LSTM, demonstrates a significant improvement over the GARCH (1,1) model, with an RMSE of 0.0430380,

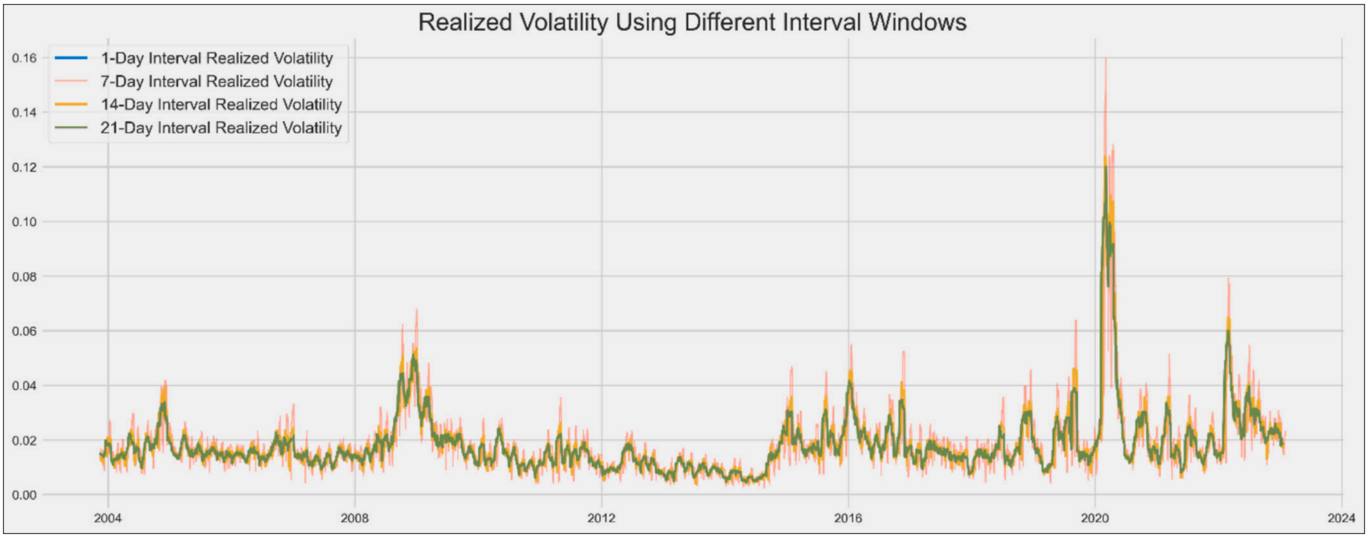

Fig. 4. Realized volatility with 1, 7, 14 and 21 steps.

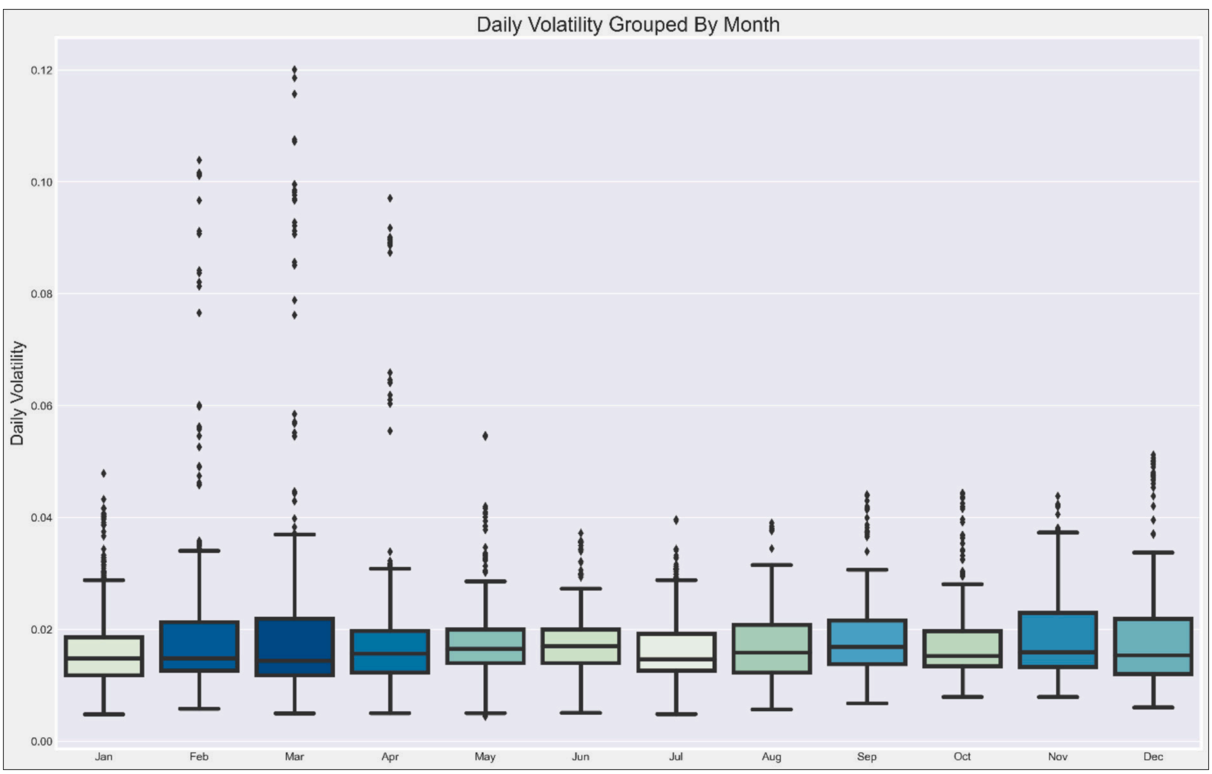

Fig. 5. Daily volatility with Interval 21 steps grouped by month.

MAPE of 0.083085, and MSE of 0.001852. These results suggest that the LSTM model provides a more accurate forecast for daily crude oil data than the GARCH (1,1) model. The GARCH (1,1) LSTM hybrid model, combining the GARCH (1,1) and LSTM techniques, further improves the forecasting performance with an RMSE of 0.0403475, MAPE of 10.342735, and MSE of 0.001627. The improvement in error values indicates that the combination of the GARCH (1,1) and LSTM approaches captures the dynamics of daily crude oil prices more effectively than using either model individually. GARCH (1,1) GRU, the final model that combines GARCH (1,1) with the GRU deep learning technique, yields the best forecasting performance among the four models, with an RMSE of 0.033079, MAPE of 6.86429, and MSE of 0.001094. This suggests that incorporating GRU into the GARCH (1,1) model leads to better results than using LSTM, providing the most accurate forecasts for daily crude oil prices.

Table 3 presents the forecasting performance of the same four models (GARCH (1,1), LSTM, GARCH (1,1) LSTM, and GARCH (1,1) GRU) on 60-min interval observations of crude oil brand prices. The performance was evaluated using the following error metrics: Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Squared Error (MSE). For the 60-min interval data, the GARCH (1,1) model yielded an RMSE of 0.1964632, MAPE of 169.847, and MSE of 0.0385697. Although these error values are lower than those observed for daily data, they still indicate a less accurate performance compared

Table 2
Results of daily observations of crude oil.

Daily observations of Crude oil			
Model	RMSE	MAPE	MSE
GARCH (1,1)	2.8896	277.82	8.34844
LSTM	0.0430380	0.083085	0.001852
GARCH (1,1) LSTM	0.0403475	10.342735	0.001627
GARCH (1,1) GRU	0.033079	6.86429	0.001094

Notes: GARCH (1,1): Generalized Autoregressive Conditional Heteroskedasticity model with parameters (1,1); GRU: Gated Recurrent Units; LSTM: Long Short-Term Memory networks; RMSE (Root Mean Square Error): A measure calculating the square root of the average squared differences between predicted and observed values. Smaller RMSE values indicate superior model fit; MAPE (Mean Absolute Percentage Error): An average of the absolute percentage differences between predicted and actual values. A reduced MAPE suggests enhanced prediction accuracy; MSE (Mean Square Error): Represents the average squared differences between predicted and true values. Lower values indicate a more accurate model.

Table 3
Results of 60 min frequency of crude oil.

60 min interval's observations of Crude oil			
Model	RMSE	MAPE	MSE
GARCH (1,1)	0.1964632	169.847	0.0385697
LSTM	0.046411	0.083474	0.002154
GARCH (1,1) LSTM	0.044542	9.58742	0.001984
GARCH (1,1) GRU	0.038691084	6.54254	0.001497

Notes: GARCH (1,1): Generalized Autoregressive Conditional Heteroskedasticity model with parameters (1,1); GRU: Gated Recurrent Units; LSTM: Long Short-Term Memory networks; RMSE (Root Mean Square Error): A measure calculating the square root of the average squared differences between predicted and observed values. Smaller RMSE values indicate superior model fit; MAPE (Mean Absolute Percentage Error): An average of the absolute percentage differences between predicted and actual values. A reduced MAPE suggests enhanced prediction accuracy; MSE (Mean Square Error): Represents the average squared differences between predicted and true values. Lower values indicate a more accurate model.

to the other models. The LSTM model's performance improved significantly with an RMSE of 0.046411, MAPE of 0.083474, and MSE of 0.002154 for the 60-min interval data. This suggests that the LSTM model is more accurate in forecasting crude oil prices at this frequency than the GARCH (1,1) model. The hybrid GARCH (1,1) LSTM model further enhances the forecasting performance with an RMSE of 0.044542, MAPE of 9.58742, and MSE of 0.001984. These improved error values indicate that the combination of GARCH (1,1) and LSTM approaches is more effective at capturing the dynamics of 60-min interval crude oil prices than using either model individually. Similar to the daily data results, the GARCH (1,1) GRU model demonstrated the best forecasting performance among the four models on the 60-min interval data, with an RMSE of 0.038691084, MAPE of 6.54254, and MSE of 0.001497. This indicates that the combination of GARCH (1,1) with.

GRU outperforms both the GARCH (1,1) LSTM and individual LSTM models.

Table 4 presents the forecasting performance of the four models (GARCH (1,1), LSTM, GARCH (1,1) LSTM, and GARCH (1,1) GRU) on 30-min interval observations of crude oil brand prices. For the 30-min interval data, the GARCH (1,1) model yielded an RMSE of 0.112605, MAPE of 218.6466, and MSE of 0.01267. Although the error values are lower than those observed for the daily and 60-min data, they still indicate less accurate performance compared to the other models. The LSTM model's performance significantly improved at this frequency, with an RMSE of 0.005353, MAPE of 0.07100, and MSE of 0.000028654930 for the 30-min interval data. This suggests that the LSTM model provides a more accurate forecast of crude oil prices at this

Table 4
Results of 30 min frequency of crude oil.

30 min interval's observations of Crude oil			
Model	RMSE	MAPE	MSE
GARCH (1,1)	0.112605	218.6466	0.01267
LSTM	0.005353	0.07100	0.000028654930
GARCH (1,1) LSTM	0.0056318	7.5727718	0.000031718
GARCH (1,1) GRU	0.0055254	7.598571	0.000032547

Notes: GARCH (1,1): Generalized Autoregressive Conditional Heteroskedasticity model with parameters (1,1); GRU: Gated Recurrent Units; LSTM: Long Short-Term Memory networks; RMSE (Root Mean Square Error): A measure calculating the square root of the average squared differences between predicted and observed values. Smaller RMSE values indicate superior model fit; MAPE (Mean Absolute Percentage Error): An average of the absolute percentage differences between predicted and actual values. A reduced MAPE suggests enhanced prediction accuracy; MSE (Mean Square Error): Represents the average squared differences between predicted and true values. Lower values indicate a more accurate model.

frequency than the GARCH (1,1) model. The hybrid GARCH (1,1) LSTM model achieves an RMSE of 0.0056318, MAPE of 7.5727718, and MSE of 0.000031718 for the 30-min interval data. These error values show a slight increase compared to the LSTM model, suggesting that the combination of GARCH (1,1) and LSTM approaches may not be as effective at capturing the dynamics of 30-min interval crude oil prices as the LSTM model alone. The GARCH (1,1) GRU model demonstrates an RMSE of 0.0055254, MAPE of 7.598571, and MSE of 0.000032547 for the 30-min interval data. These error values are slightly higher than those of the LSTM model, indicating that the GARCH (1,1) GRU model is less accurate than the LSTM model at this frequency.

Table 5 & 6 show the results for the 1-min frequency for samples I and II. The LSTM model had the lowest RMSE and MAPE, indicating that it had the best overall predictive accuracy among the models. However, the GARCH (1,1)-GRU hybrid model has the lowest MSE, implying that it has the best performance in terms of minimizing squared errors. The GARCH (1,1) model has the highest error values for all three metrics, indicating that it is the least accurate model for predicting crude oil prices, similar to the results for Sample I. Furthermore, the LSTM model has the lowest RMSE and MAPE, indicating that it has the best overall predictive accuracy among the models. The GARCH (1,1)-GRU hybrid model has the lowest MSE, implying that it has the best performance in terms of minimizing squared errors. The GARCH (1,1) model has the highest error values for all three metrics, indicating that it is the least accurate model for predicting crude oil prices. In conclusion, the LSTM and GARCH (1,1)-GRU hybrid models appear to be the best choices for predicting crude oil prices based on 1-min interval observations for Sample II, with the LSTM model having a slight edge in overall

Table 5
Results of 1 min frequency of crude oil- Sample I.

Sample I: 1 min interval's observations of Crude oil			
Model	RMSE	MAPE	MSE
GARCH (1,1)	0.03926502	417.266439	0.001541742
LSTM	0.000794	0.0389407	0.0000006
GARCH (1,1) LSTM	0.00071859	4.2724151	0.00000051638
GARCH (1,1) GRU	0.00062463	4.18254	0.00000035423

Notes: GARCH (1,1): Generalized Autoregressive Conditional Heteroskedasticity model with parameters (1,1); GRU: Gated Recurrent Units; LSTM: Long Short-Term Memory networks; RMSE (Root Mean Square Error): A measure calculating the square root of the average squared differences between predicted and observed values. Smaller RMSE values indicate superior model fit; MAPE (Mean Absolute Percentage Error): An average of the absolute percentage differences between predicted and actual values. A reduced MAPE suggests enhanced prediction accuracy; MSE (Mean Square Error): Represents the average squared differences between predicted and true values. Lower values indicate a more accurate model.

Table 6
Results of 1 min frequency of crude oil-Sample II.

Sample II: 1 min interval's observations of Crude oil			
Model	RMSE	MAPE	MSE
GARCH (1,1)	0.095406	399.60521	0.0091023
LSTM	0.001698	0.03191	0.000002884647
GARCH (1,1) LSTM	0.0007287	4.29541	0.0000005768
GARCH (1,1) GRU	0.000619863	4.1978	0.00000038423

Notes: GARCH (1,1): Generalized Autoregressive Conditional Heteroskedasticity model with parameters (1,1); GRU: Gated Recurrent Units; LSTM: Long Short-Term Memory networks; RMSE (Root Mean Square Error): A measure calculating the square root of the average squared differences between predicted and observed values. Smaller RMSE values indicate superior model fit; MAPE (Mean Absolute Percentage Error): An average of the absolute percentage differences between predicted and actual values. A reduced MAPE suggests enhanced prediction accuracy; MSE (Mean Square Error): Represents the average squared differences between predicted and true values. Lower values indicate a more accurate model.

predictive accuracy, whereas the GARCH (1,1)-GRU hybrid model performs better in minimizing squared errors.

For daily, 60-min, and 30-min interval observations, the LSTM model had the lowest RMSE and MAPE values, indicating a better overall predictive accuracy. The GARCH (1,1)-GRU hybrid model had the lowest MSE values for these time intervals, indicating better performance in minimizing squared errors. For both 1-min interval samples, the LSTM model had the lowest RMSE and MAPE values, whereas the GARCH (1,1)-GRU hybrid model had the lowest MSE values. In conclusion, the LSTM and GARCH (1,1)-GRU hybrid models consistently outperform the GARCH (1,1) and GARCH (1,1)-LSTM hybrid models in predicting crude oil prices across various time intervals. The LSTM model has a slight edge in the overall predictive accuracy, whereas the GARCH (1,1)-GRU hybrid model performs better in minimizing the squared errors. Based on the results of this study, we contribute to theoretical, empirical, and marginal economic effects. (i) Theoretical Economic Benefits: We present a study with theoretical and economic implications. Through our comparison of forecasting models, we add to the existing literature on predictive modeling of financial markets. The noteworthy performance of the LSTM and GARCH (1,1)-GRU hybrid models indicates that incorporating deep learning techniques alongside traditional time-series models can significantly improve predictive accuracy. This finding has theoretical implications for the integration of advanced machine learning methods into financial modeling, which has the potential to result in more accurate predictions in various economic contexts; (ii) Empirical Economic Benefits: Based on empirical evidence, our research offers practical implications for practitioners and policymakers. The validity of our models in forecasting crude oil prices across various time horizons has significant implications for investment and risk-management decisions in the energy and financial sectors. Enhanced forecasting accuracy allows businesses to make informed decisions regarding resource allocation, thereby minimizing the probability of financial losses. Furthermore, policymakers can leverage more accurate predictions to formulate effective energy and economic policies, ultimately leading to improved economic stability; and (iii) Marginal Economic Effects (Marginality Analysis): In light of the marginal economic effects, our research reveals that incorporating advanced deep learning models, particularly the GARCH (1,1)-GRU hybrid, can result in substantial cost savings and risk reduction. The decreased errors, as demonstrated by the lower MSE values, signify that decision makers can anticipate more accurate forecasts, resulting in more efficient resource allocation and better risk mitigation. More accurate forecasts lead to tangible cost savings because businesses can optimize their operations and financial strategies more effectively.

5. Conclusion and policy implications

This study conducted an empirical analysis of Brent crude oil prices using various time frequencies and sample periods to investigate the forecasting performance of different models, including GARCH (1,1), Vanilla LSTM, GARCH (1,1) LSTM, and GARCH (1,1) GRU, on crude oil price data. This study used daily and intraday observations of crude oil prices, and the data consisted of different sample periods for each observation frequency. The results demonstrate that the LSTM and GARCH (1,1)-GRU hybrid models outperformed the other models, with the LSTM model having a slight edge in overall predictive accuracy, whereas the GARCH (1,1)-GRU hybrid model performed better in minimizing squared errors.

The outcomes of this study have substantial implications for the global energy market and manufacturing sectors that depend on crude oil prices. Economic sustainability relies on the successful co-movement of the expansion of energy and manufacturing sectors. As crude oil is a raw material with a significant influence, market failure, uncertainty, and negative externalities in the fossil fuel market can severely affect the global economic system through price volatility. To limit volatility, this study underscores the significance of accurately forecasting crude oil prices, which is critical for risk management and the stability of the global economic system. The depletion of energy resources is inevitable, and it can unleash an energy market that may become more volatile in the future owing to negative environmental externalities, market asymmetry, structural changes in the economy, and unregulated market conditions. Crude oil price stability predominantly affects the economic performance of developed and emerging economies. Thus, volatility in the crude oil market can be deleterious to businesses and manufacturing sectors across these economies, disrupting the production and distribution chains of goods and services.

The consequences of a volatile crude oil market can be catastrophic for firms and businesses that cannot cope with sudden price changes. They are forced to switch to alternative energy sources (e.g., coal), which have far-reaching negative environmental impacts. Those who cannot switch to crude oil substitutes must close down, leading to job losses, recessions, and economic policy uncertainties. Inaccurate forecasting can induce energy market volatility, leading to political instability and unexpected regime change. The findings of this study offer valuable insights into the use of advanced machine learning techniques in crude oil price forecasting, enabling practitioners to improve the accuracy of their predictions and make informed decisions. Investors rely on accurate forecasts to make investment decisions on the crude oil market, making these insights particularly valuable. Furthermore, the study's implications are relevant to researchers in the field of finance, providing an empirical basis for further research on the use of advanced machine learning techniques for crude oil price forecasting. By exploring different sample periods and time frequencies, researchers can gain a deeper understanding of crude oil price behavior and develop more accurate forecasting models. Given the ongoing conflict between Russia and Ukraine and its significant impact on global markets, our study and its findings are critically important. The region's dependence on Russia as a supplier of oil and gas to European countries renders it vulnerable to energy market shocks that result from geopolitical tensions. Our refined forecasting models for crude oil prices serve as a beacon of predictability, helping mitigate these shocks and provide stability. This predictive accuracy enables countries, particularly those heavily reliant on Russian energy, to proactively seek alternative energy sources or strategically manage their reserves. Moreover, these forecasts function as bulwarks against global financial market volatility, offering institutions a chance to brace for and hedge against potential upheavals. This foresight becomes an invaluable asset for governments, guiding them in formulating policies related to energy, trade, and diplomacy, and equipping them to diversify energy partnerships or potentially mediate conflicts. Businesses, especially those entrenched in regions sensitive to Russia-Ukraine dynamics, can leverage this information for supply chain adjustments, risk management, and continuity. As the world struggles to recover from COVID-19, the ability to accurately predict oil prices adds a layer of stability, ensuring that energy price surges and geopolitical frictions do not derail. Furthermore, the enhanced forecasting capability not only mitigates potential market disasters, but also works toward building an environment of trust and predictability among global stakeholders. In essence, while our study focuses on crude oil price forecasting, its broader implications offer a roadmap for navigating the complex economic challenges intertwined with the Russia-Ukraine tensions and post-COVID recovery landscape. Policymakers are urged to consider maintaining strategic reserves of crude oil as a measure to counter supply disruptions and price volatility, particularly during times of crisis. These reserves play a crucial role in stabilizing prices and ensuring energy security for the country. Additionally, encouraging the diversification of energy sources is recommended as a strategy to reduce reliance on crude oil and mitigate the impact of price fluctuations. This diversification effort can involve promoting renewable energy sources like solar, wind, and hydroelectric power. Investing in energy efficiency is highlighted as another vital policy approach to reducing overall energy consumption and mitigating the impact of price increases. This can be achieved through incentives for energy-efficient technologies and practices in both industries and households. Furthermore, enhancing market transparency in the crude oil market is crucial for reducing uncertainty and price volatility. Policymakers are encouraged to collaborate with industry stakeholders to improve reporting standards and facilitate market information sharing. Promoting sustainable practices in the energy and manufacturing sectors is emphasized as a means to reduce environmental externalities and contribute to long-term economic stability. This can include incentivizing cleaner production processes and emissions reduction. Lastly, developing and implementing robust risk management and contingency plans is essential for mitigating the impact of price volatility. Policymakers can consider implementing hedging strategies and emergency response plans to manage these risks effectively. Overall, this study underscores the importance of incorporating advanced machine learning techniques in crude oil price forecasting. The crude oil market is complex and volatile, and traditional models may fail to capture the underlying patterns and dynamics of crude oil prices. Advanced machine-learning techniques enable practitioners and researchers to improve the accuracy of their predictions, make more informed decisions, and ultimately contribute to the stability of the global economic system.

Although our study conveys novel findings regarding forecasting of one of the world's most traded commodity markets, there are some limitations that need to be acknowledged. This is instrumental in instigating future research in this domain to better grasp the pattern of the crude oil market, especially during turbulent periods. We encountered two constraints related to data mining and execution. Because forecasting accuracy largely depends on the size of the data, it is instrumental to work with high-frequency datasets and datasets that can represent the studied commodity markets. The unavailability of high-frequency datasets was a key limitation of this study. Moreover, big data and advanced deep learning techniques can be complementary in the sense that conventional data-handling approaches have limited forecasting capabilities, yielding a flimsy connection between big data and conventional machine-learning approaches. Consequently, future studies can investigate how advanced deep learning approaches can be useful tools for handling big datasets. In addition, future studies can investigate further simulation-based studies combining the effects of exogenous economic shocks to better predict the fluctuations of market returns related to critical commodity markets, such as crude oil. This study acknowledges several limitations that pave the way for future research. Firstly, the predictive models used, while effective, may benefit from the inclusion of additional macroeconomic variables such as interest rates, inflation rates, and GDP growth, which could further enhance forecasting accuracy. The inclusion of these variables could provide a more comprehensive understanding of the factors influencing crude oil prices. Secondly, the study focuses on Brent crude oil prices; future research could extend the analysis to other types of crude oil, such as WTI or Dubai crude, or even to other energy commodities like natural gas or coal, to validate the generalizability of the findings across different markets. Moreover, the research could be enriched by exploring the impact of geopolitical events, such as conflicts, sanctions, or trade agreements, on crude oil price volatility. Incorporating sentiment analysis from news and social media could provide insights into how market perceptions and investor sentiment influence price movements. Additionally, the application of emerging machine learning techniques, such as deep reinforcement learning or neural architecture search, could provide novel insights into crude oil price forecasting by potentially uncovering complex patterns and relationships in the data that traditional models may miss. Finally, considering the evolving nature of the global energy market, future studies should continuously update and refine the models to adapt to new market dynamics and technological advancements. This could involve incorporating real-time data feeds, exploring the impact of renewable energy sources on oil demand, or examining the effects of climate change policies on oil markets. By addressing these limitations and exploring these future research directions, the field can continue to advance in its understanding and prediction of crude oil price movements.

CRediT authorship contribution statement

Amar Rao: Methodology, Investigation, Formal analysis, Data curation. **Gagan Deep Sharma:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aviral Kumar Tiwari:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Razib Hossain:** Writing – review & editing, Writing – original draft. **Dhairya Dev:** Writing – review & editing, Writing – original draft.

Appendix A

GARCH (1,1): The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model (1,1) is a fundamental tool in time series analysis, particularly in the realm of financial econometrics. It was introduced by Robert F. Engle in the early 1980s as a means to address the inherent volatility clustering observed in financial markets. The GARCH (1,1) model is widely used for analyzing and forecasting volatility, providing valuable insights into the dynamics of asset returns.

At its core, the GARCH (1,1) model builds upon the principles of autoregressive conditional heteroskedasticity (ARCH), which was proposed by Robert F. Engle in 1982. The ARCH model assumes that the conditional variance of a time series is a function of past squared observations, implying that volatility clusters in the data. However, the ARCH model does not account for the persistence of volatility over time. To address this limitation, the GARCH model extends the ARCH framework by incorporating past conditional variances into the model. Specifically, the GARCH (1,1) model includes autoregressive and moving average components to capture the dynamics of volatility. The term "1,1" signifies the order of these components in the model: one lag for the autoregressive component and one lag for the moving average component.

Mathematically, the conditional variance in a GARCH (1,1) model can be expressed as:

$$\left[\sigma_t^2 = \omega + \alpha\epsilon_{\{t-1\}}^2 + \beta\sigma_{\{t-1\}]}^2 \right.$$

where :

$-\left(\sigma_t^2 \right)$ *represents the conditional variance of the time series at time* (t),

$-(\omega)$ *is the constant term representing the long* $-$ *term average variance*,

$-(\alpha)$ *and* (β) *are parameters that govern the persistence and decay of volatility*, *respectively*,

$-\left(\epsilon_{\{t-1\})}^2 \right.$*is the squared residual at time* $(t-1)$.

The GARCH (1,1) model is highly relevant for forecasting accuracy in crude oil prices due to its ability to capture volatility clustering, a phenomenon commonly observed in financial markets. Crude oil prices are known for their inherent volatility, influenced by a myriad of factors such as geopolitical events, supply-demand dynamics, and macroeconomic indicators. By incorporating past volatility information into the forecasting process, the GARCH (1,1) model can effectively capture the persistence and clustering of volatility in crude oil prices. This enables analysts and policymakers to better understand and anticipate changes in market dynamics, facilitating more informed decision-making and risk management strategies. Furthermore, the GARCH (1,1) model provides valuable insights into the underlying structure of volatility in crude oil markets, shedding light on the factors driving price fluctuations. This knowledge can inform the development of more robust forecasting models and trading strategies, enhancing overall market efficiency and stability. In summary, the GARCH (1,1) model serves as a powerful tool for analyzing and forecasting volatility in crude oil prices, offering valuable insights into market dynamics and risk management. Its relevance in the realm of financial econometrics underscores its importance as a cornerstone of modern time series analysis.

Vanilla LSTM: Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) architecture that has gained popularity for its ability to model sequential data effectively. It was introduced by Sepp Hochreiter and Jürgen Schmidhuber in 1997 as a solution to the vanishing gradient problem, which is commonly encountered in training traditional RNNs. Vanilla LSTM refers to the basic LSTM architecture without any additional modifications or enhancements. The key innovation of LSTM lies in its architecture, which includes a set of gates that regulate the flow of information through the network. These gates, including the input gate, forget gate, and output gate, control the information flow in and out of the memory cell, allowing LSTM networks to retain information over long sequences and avoid the vanishing gradient problem.

Mathematically, the operations of an LSTM cell can be summarized as follows:

$$\left[f_t = \sigma_{g\left(W_f \cdot \left[h_{\{t-1\}}, x_t \right] + b_f \right)} \right]$$

$$\left[i_t = \sigma_{g\left(W_i \cdot \left[h_{\{t-1\}}, x_t \right] + b_i \right)} \right]$$

$$\left[\widetilde{\{C\}}_t = \backslash tanh\left(W_C \cdot \left[h_{\{t-1\}}, x_t \right] + b_C \right) \right]$$

$$\left[C_t = f_t * C_{\{t-1\}} + i_t * \widetilde{\{C\}}_t \right]$$

$$\left[o_t = \sigma_{g\left(W_o \cdot \left[h_{\{t-1\}}, x_t \right] + b_o \right)} \right]$$

$$[h_t = o_t * \backslash tanh(C_t)]$$

where :

$-(f_t)$ *is the forget gate output*,

$-(i_t)$ *is the input gate output*,

$-\left(\tilde{}\{C\}_t \right)$ *is the candidate cell state*,

$-(C_t)$ *is the cell state*,

$-(o_t)$ *is the output gate output*,

$-(h_t)$ *is the hidden state*,

$-(x_t)$ *is the input at time* (t),

$-\left(\sigma_g \right)$ *is the sigmoid activation function*,

$-\left(W_f, W_i, W_C, W_o \right)$ *are the weight matrices*,

$-\left(b_f, b_i, b_C, b_o \right)$ *are the bias vectors*.

Vanilla LSTM is highly relevant for forecasting accuracy in crude oil prices due to its ability to capture long-term dependencies and patterns in the data. Crude oil prices exhibit complex dynamics influenced by various factors, including geopolitical events, economic indicators, and supply-demand dynamics. Traditional time series models may struggle to capture these intricate relationships, leading to suboptimal forecasts. By leveraging its memory cell and gating mechanisms, Vanilla LSTM can effectively capture and retain information over long sequences, enabling it to identify and exploit long-term patterns in crude oil price data. This ability is crucial for predicting price movements, as it allows the model to learn from past observations and adapt to changing market conditions. Furthermore, Vanilla LSTM's ability to overcome the vanishing gradient problem ensures that it can be trained effectively on large datasets, further enhancing its forecasting accuracy. This makes it a valuable tool for analysts and researchers seeking to improve the accuracy of their crude oil price forecasts and gain deeper insights into market dynamics.

GARCH (1,1) GRU: The GARCH (1,1) Gated Recurrent Unit (GRU) model is a novel approach that combines the GARCH (1,1) model with the GRU architecture, a type of recurrent neural network (RNN). This hybrid model leverages the strengths of both GARCH and GRU to enhance the forecasting accuracy of crude oil prices. The GARCH (1,1) model, as discussed earlier, is a time series model used to analyze and forecast volatility in financial markets. It captures the volatility clustering often observed in financial data by incorporating autoregressive and moving average components. On the other hand, GRUs are a simplified version of LSTM networks that also address the vanishing gradient problem. GRUs have fewer parameters and are computationally less expensive than LSTMs, making them more efficient for training on large datasets. Despite their simplicity, GRUs can still capture long-term dependencies in sequential data, making them suitable for modeling time series data like crude oil prices.

The conditional variance in a GARCH (1,1) model is given by:

$$\left[\ \sigma_t^2 = \omega + \alpha\epsilon^2_{\{t-1\}} + \beta\sigma^2_{\{t-1\}]}\right.$$

where :

$-\left(\ \sigma_t^2\right)$ *is the conditional variance at time* $(\ t\ )$,

$-(\ \omega)$ *is the constant term representing the long – term average variance*,

$-(\ \alpha)$ *and* $(\ \beta)$ *are parameters that govern the persistence and decay of volatility*, *respectively*,

$-\left(\ \epsilon^2_{\{t-1\})}\right.$*is the squared residual at time* $(\ t-1\ )$.*The update gate* $(\ z_t)$, *reset gate* $(\ r_t)$, *and new memory content* $\left(\ \widetilde{\{h\}}_t\right)$ *in a GRU cell*

are computed as follows :

$$[\ z_t = \sigma\left(W_z\cdot\left[h_{\{t-1\}}\ , x_t\right]\ \right)\ ]$$

$$[\ r_t = \sigma\left(W_r\cdot\left[h_{\{t-1\}}\ , x_t\right]\ \right)\ ]$$

$$[\ \{h\}_t \backslash tanh\left(W\cdot\left[r_t \odot h_{\{t-1\}}\ , x_t\right]\ \right)\ ]$$

where :

$-(\ z_t)$ *is the update gate at time* $(\ t\ )$,

$-(\ r_t)$ *is the reset gate at time* $(\ t\ )$,

$-\left(\ \{h\}_t\right.$*is the new memory content at time* $(\ t\ )$,

$-(\ \sigma)$ *is the sigmoid activation function*,

$-(\ \odot\ )$ *denotes element – wise multiplication*,

$-(\ W_z), (\ W_r)$, *and* $(\ W\ )$ *are the weight matrices*.

The GARCH $(1,1)$*GRU model combines the GARCH* $(1,1)$*model with the GRU architecture by using the GRU's output* $(\ h_t)$

to model the volatility :

$$\left[\ \sigma_t^2 = \omega + \alpha\epsilon^2_{\{t-1\}} + \beta\sigma^2_{\{t-1\}} + \gamma h_t\right]$$

where :

$-(\ \gamma)$ *is a parameter that governs the impact of the GRU output on volatility*.

This combined model allows the GARCH (1,1) GRU model to capture both volatility patterns and sequential dependencies in the data, making it relevant for forecasting accuracy in crude oil prices. The combination of GARCH and GRU in the GARCH (1,1) GRU model offers several advantages for forecasting crude oil prices. By integrating the GARCH framework, the model can capture volatility patterns in the data, which are essential for predicting price movements in volatile markets like crude oil. Additionally, the GRU component allows the model to capture long-term dependencies in the data, further enhancing its forecasting accuracy. This combination of volatility modeling and sequential data analysis enables the GARCH (1,1) GRU model to capture both the short-term fluctuations and long-term trends in crude oil prices, making it a powerful tool for analysts and researchers in the energy sector. Overall, the GARCH (1,1) GRU model represents a sophisticated approach to forecasting crude oil prices, leveraging the complementary strengths of GARCH and GRU to improve forecasting accuracy and capture the complex dynamics of crude oil markets.

Table A1

Daily			
ADF		PP	
Level	1st difference	Level	1st difference
t-Statistic	t-Statistic	Adj. t-Stat	Adj. t-Stat
−2.08959	−88.5477***	−2.10078	−88.5814***
Weekly			
−2.08	−20.321***	−2.332	−38.8315***

Note: ADF- Augmented Dickey-Fuller test statistic, PP-Phillips-Perron test statistic.
*** p-value<0.01.

Data availability

Data will be made available on request.

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Amar Rao

Dr. Amar Rao is an Associate Professor at Shoolini University and a researcher in finance and AI. He has published in high-impact journals including Renewable Energy, Resources Policy, Global Finance Journal, Technology Forecasting and Social Change, and Annals of Operations Research. He is an expert in AI, machine learning, finance, and the environment. He is working on an incubation hub and regularly participates in conferences and faculty development programs, serving as a resource person for AI and research methods.

Gagan Deep Sharma

Gagan Deep Sharma currently works as a Professor in Management at Guru Gobind Singh Indraprastha University, New Delhi, India. He is Associate Director, Office of International Affairs of the University. He holds PhD in Management and master's in philosophy and commerce. He has an academic experience of over twenty years. He has published in, reviewed for, and edited/guest-edited the journals of high impact.

Aviral Kumar Tiwari

Dr. Aviral Kumar Tiwari is the Chairperson of Research and Publications at the Indian Institute of Management Bodh Gaya (IIMBG). Prof. Tiwari is a C-EENRG fellow at the Department of Land Economy, University of Cambridge and Research Fellows, University of Economics Ho Chi Minh City, Vietnam. Prior to joining IIMBG, he worked as Associate Professor at Rajagiri Business School (RBS), India and Montpellier Business School (MBS), Montpellier, France, from where he received his post-doc as well. After graduating with a degree in Economics from Lucknow University, Lucknow, India, he received his M.Phil. (in Labour Economics) and PhD (in Energy and Environment) from ICFAI University Tripura. He is basically an applied economist with broad empirical interests with a focus on but not limited to emerging economies, in particular Asia. His research interests focus on various issues concerning energy, environment, tourism, macroeconomy and growth & development etc. He has published widely in peer-reviewed international journals and contributed >100 ABDC-A &A* research papers so far. He is the only Economist included from India in the career-long list of the World's top 2 % scientists of 2021 published by Stanford University Study. He is consistently ranked at the top position in India as a researcher by IDEAS/PePEc. He is one of the Highly Cited Researchers 2020 by Clarivate™ of Web of Science™. He is one of the recipients of the M J Manohar Rao Award (for young econometricians) for 2014 from The Indian Econometric Society, 2015. He is a Life member, of the Indian Economic Association, of India and a member of several other international associations such as the International Association for Energy Economics (IAEE), USA, Association for Comparative Economic Studies, USA, Western Economic Association International, USA etc. He holds different editorial positions at >10 journals of international repute (ABDC, Scopus and ABS-indexed journals).

Mohammad Razib Hossain

Mohammad Razib Hossain is a Doctoral Researcher at the University of Adelaide, Australia. Before joining as a full-time researcher, he has accomplished his MSc (Economics) degree from the University of Leeds, United Kingdom. He is a Commonwealth Scholar at the University of Leeds. He received many awards and recognitions including the Prime Minister Gold Medal and the University Award due to his academic excellence. His research interests lie within Ecological Economics, Energy Economics, Renewable Energy, Time-series Econometrics, Panel data Modeling, Circular Economy, and Carbon Mitigation. He has published his works in journals like Energy Economics, Energy Policy, Resources Policy and so on.

Dhairya Dev

Dhairya Dev is a doctoral student at the University School of Management Studies at Guru Gobind Singh Indraprastha University. His research area is "sustainable investment", and he focuses primarily on identifying, and rating investments. Also, encourage investors to invest in sustainable investment avenues and generate returns. He also aims to study the role of governance and strengthen the policy of sustainable investment. Other interests: sustainable development, CSR, ESG, and systematic reviews.